EMR-MR Sudan — Software Requirements Specification

Oct 4, 2026

1. Purpose and scope

This document lists what the EMR-MR system must do at launch. There is no pilot: the system goes live directly, with detailed areas for Khartoum and an area map covering all of Sudan. It covers every step of an emergency: a person calls the emergency number, the Sudan Health Authority call centre sends a request to the nearest hospital, the hospital sends its own ambulance, the crew treats the patient, the hospital receives them, and the case is closed. It is based on the Ibrahim prototype (34 screens) and our gap analysis. From now on, the team builds from this document, not from the prototype.

In the first release: taking emergency calls, deciding how urgent each call is, sending ambulances, tracking ambulances on the map, the paramedic's patient report, handing the patient to the hospital, hospital beds, the basic patient record, user and system settings, and performance reports.

Later: the full medical record (medicines, lab tests, X-rays and scans, operations, vaccines, visits), reports to the Ministry's DHIS2 system, checking national ID, links to hospital lab systems, large accidents with many patients, and sharing cases with police and civil defence.

Priority words: Must = needed before launch. Should = add in the first release if there is time, otherwise in the next release. Could = later.

Language: the system works fully in both Arabic and English, as equal languages. Each user chooses their language; Arabic screens are right-to-left and English screens are left-to-right.

Critical areas (read first)

These eight areas are the most important. If one of them fails, people can die. Each one must pass a real-life test and a practice drill before launch. Any change to them needs approval from doctors and operations managers.

#	Area	Why it matters	Must work at launch	Requirements
1	Taking the call and finding the place	If we do not know where the patient is, no ambulance can reach them. Many places in Sudan have no street address.	Every case has an area name plus a map point or a landmark. Repeat calls about the same case are found.	CTD-01, CTD-02, CTD-04, BR-02
2	Deciding how urgent the call is	Without this, a heart attack can wait behind a small injury.	The system gives every case a level from P1 (most urgent) to P4 automatically, and sorts all lists by it. The system warns if a P1 case waits more than 60 seconds.	CTD-03, CTD-06, BR-01
3	Sending to the right hospital	Sending to a closed hospital, or one with no free ambulance, wastes time.	Only working hospitals with a free ambulance are suggested. If a hospital does not answer in 5 minutes, the call centre sends the request to the next one.	CTD-07 to CTD-09, AMD-01, AMD-06, BR-03, BR-05

#	Area	Why it matters	Must work at launch	Requirements
4	Tracking and working without internet	The network often goes down. The work must not stop.	The ambulance apps work with no internet. If the app does not confirm a job, the hospital phones the crew and records the call. The route is saved and sent later.	AMD-03 to AMD-05, CTD-08, NFR-05, NFR-06
5	Handing the patient to the hospital	If the hospital is not ready, the ambulance waits at the door and cannot go to the next call.	The hospital gets a summary before the patient arrives, says yes or no, and confirms the handover.	AMS-06, HER-01 to HER-04, BR-07
6	Correct patient information	Wrong vital signs, missing allergies or mixed-up patients can harm the patient.	Vital signs have a time and a name. Allergies are always shown. Records are never deleted.	AMS-03, AMS-05, HST-01, HST-07, BR-08
7	Security and access	Leaking national health records hurts patients and trust.	Each user sees only their own data. Login needs a second code sent by SMS. Every access is recorded.	ADM-01, ADM-05, BR-09, NFR-07 to NFR-10
8	System always on, with a backup plan	If the system stops, calls and ambulances stop.	The system works 99.9% of the time, has a backup site, and staff have practised a paper-and-radio plan.	NFR-01, NFR-02, NFR-16

Client decisions (October 2026)

The client answered the open questions on blockers, gaps and critical areas. These decisions replace earlier assumptions in this document.

#	Topic	Decision	What it changes
1	Emergency number	999; the caller's number shows on the agent's screen	The call centre needs a phone link that passes the caller's number
2	Reaching ambulance crews	Ambulances have internet. The app must confirm each job; if it is not confirmed, the hospital phones the crew and the call is recorded in the system	Replaces the SMS job backup with a recorded phone call
3	Language	Arabic and English	Both languages on every screen
4	Login	Second login code by SMS only	SMS sent through Zain
5	Call centre	One or more agents per shift	Several agents can work at the same time
6	Locations	Areas for all of Sudan, in fine detail for Khartoum (no official street names). The agent drops a pin; the system shows the area around it (300 m by default, adjustable)	Building the area map is an early, key task
7	Urgency questions	A standard triage model	No need to write our own question set
8	Hospital answer time	5 minutes before moving to the next hospital; the answer is recorded by the hospital emergency role	Timer set to 5 minutes. Recommendation: 2 minutes for P1 life-threatening cases [confirm]

#	Topic	Decision	What it changes
9	Ambulances	Each hospital has one or more and adds them in the system	Hospitals manage their own ambulances
10	First-time record	All first health checks, plus each hospital's recommended checks	Each hospital can set its own extra checks
11	Patient ID	National number or driving licence	Two ID types in the patient record
12	Hospital status and beds	Updated by hospital staff in the system	No change
13	Hosting	Amazon Web Services (AWS)	Use the nearest AWS region; Health Authority approval for data stored outside Sudan [confirm]
14	Backup power and paper plan	Available	No change
15	Connections	No DHIS2 access yet; SMS by Zain	DHIS2 moves to a later release
16	Devices	Bought by the hospitals	No change
17	Training and support	Our company team	We run training and the help desk
18	Pilot	No pilot; the system goes live directly	"Pilot" is replaced by "first release"

Still open: who approves the patient-data rules (Health Authority, Ministry, or both).

Words used in this document

Word	What it means
Portal	A set of web screens for one type of user.

Word	What it means
P1 to P4	How urgent a case is: P1 = life in danger, P4 = not urgent.
Dispatch	Choosing and sending an ambulance.
Handover	The crew passes the patient to hospital staff.
Must / Should / Could	Must = needed before launch; Should = add if there is time; Could = later.
Offline	Working without internet; the information is sent later.
GPS	The phone's location, used to track ambulances.
DHIS2	The Ministry's national reporting system. It receives monthly totals, not patient names.
SMS	Text message.
FHIR	An international standard way for health systems to share information.
ICD-10	The world list of disease codes.
UTC	World time. The system saves times in world time and shows them in Sudan time.
Uptime	The share of time the system is working.

2. Users and portals

There are six user types, each with its own screens. Each user sees only the data for their own hospital, ambulance or state. The emergency call centre is run by the **Sudan Health Authority**, and the ambulances belong to the hospitals.

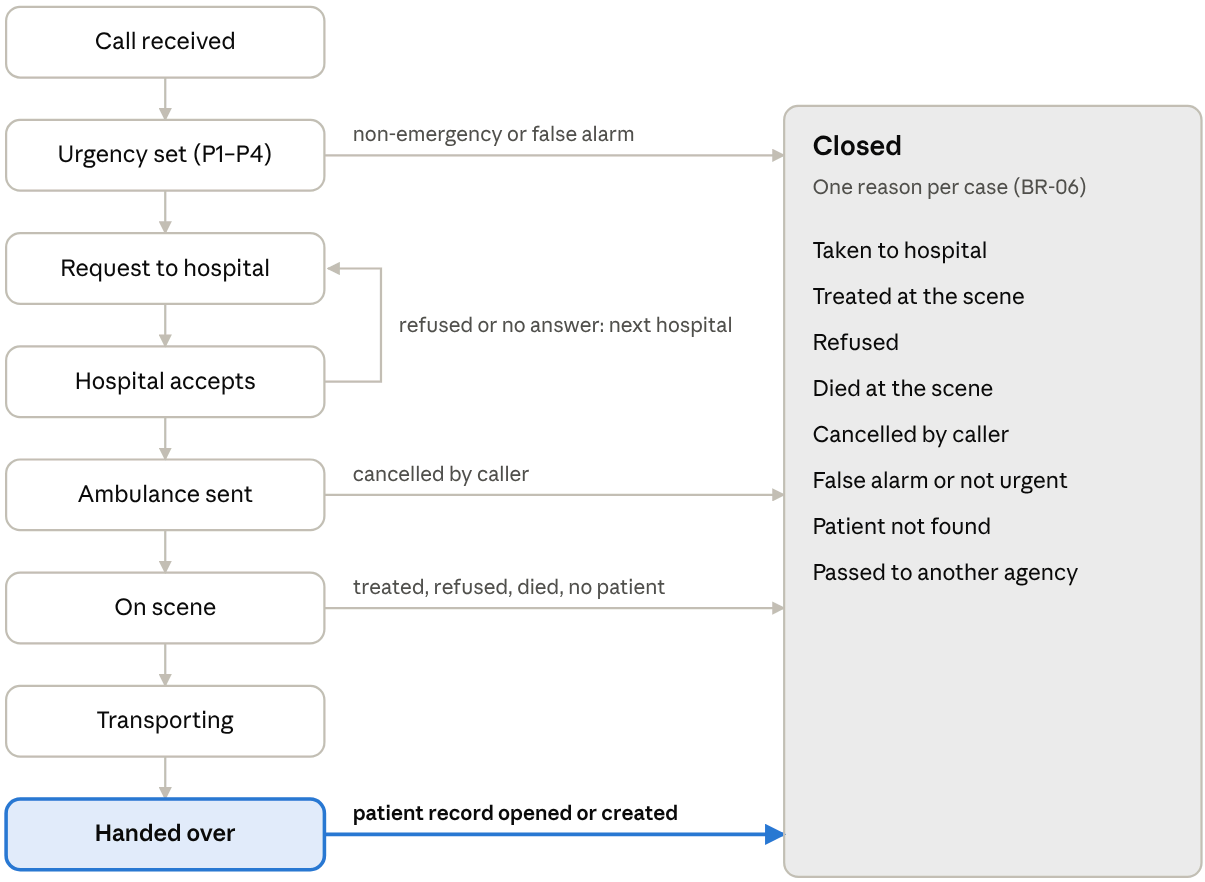
Code	User type	Who	Device	What they do
CTD	Health Authority call centre	Sudan Health Authority call-centre staff	Computer with headset	Answer emergency calls, decide how urgent they are, see the nearby hospitals on a map, send a request to the nearest suitable hospital, watch all open cases
HER	Hospital emergency staff	Emergency room doctors and nurses	Computer or tablet	Receive the request from the call centre, accept it, send one of the hospital's own ambulances, receive the patient
AMD	Ambulance driver	Hospital ambulance drivers	Android phone in the vehicle	Accept the job, follow the map, share location, press status buttons
AMS	Ambulance staff	Hospital paramedics	Android phone or tablet	Find or add the patient, record vital signs and treatment, send a summary to the hospital
HST	Hospital staff	Ward doctors, nurses, records and lab staff	Computer	Keep the patient record. For a first-time patient, create a full new record with height, weight, blood type and all required health checks.
ADM	System admin	Health Authority IT staff	Computer	Add users, set up lists (hospitals, areas, ambulances), settings and reports. Does not handle live calls.

Keeping call handling and system setup in separate accounts means call-centre staff cannot change users or settings by mistake. [confirm the official name of the Health Authority]

3. Case lifecycle

Every case goes through up to eight steps and always ends as Closed, with one reason. If a hospital refuses or does not answer, the case goes back and the call centre sends it to the next hospital.

The Health Authority sends each request to a hospital; the hospital sends its ambulance



case lifecycle · 8 states, 4 exits to closure

Who moves the case forward: the Health Authority call centre for the first three steps; hospital emergency staff accept the case and send an ambulance; the driver's app from "Ambulance sent" to "Transporting"; hospital staff for "Handed over". At handover, the

hospital opens the patient's record, or creates a full new record if this is the patient's first time.

4. Functional requirements by portal

There are 49 requirements. 40 are Must (needed before launch). Each one has a test ("How we check") that the client will use to accept the system. Times like "60 seconds" are starting values and can be changed.

4.1 Health Authority call centre (CTD)

ID	What the system must do	How we check	Priority
CTD-01	Open a new case when a call is answered, with the caller's number, the time and the call-taker's name	The case opens in 1 second with the number already filled in	Must
CTD-02	Save where the patient is: the agent drops a pin on the map and the system shows the area around it (300 m by default) with the area name from a detailed list (there are no official street names), plus landmark, written directions, and the caller's phone location by SMS link if possible	The ambulance cannot be sent until there is an area name plus a map point or landmark	Must
CTD-03	Work out the urgency automatically (P1 most urgent to P4) with a built-in priority function, from the agent's answers to the standard triage questions; the agent can change it only with a written reason	The same answers always give the same P level; every manual change is recorded with the reason	Must
CTD-04	Warn when the same emergency is called in twice	Same number, or a place within 500 m, in the last 30 minutes shows a warning	Must

ID	What the system must do	How we check	Priority
CTD-05	Close calls that are not emergencies, cut off, or fake	Every closed call has a reason from a list	Must
CTD-06	Sort open cases automatically: highest priority first, then longest waiting; the list re-sorts by itself when a priority changes	A P1 case waiting more than 60 seconds turns red	Must
CTD-07	Suggest the nearest working hospitals that have a free ambulance	Top 3 hospitals shown with distance and free ambulances; choosing another needs a reason	Must
CTD-08	Send the case to the chosen hospital; the hospital must accept	No answer in 5 minutes warns the call centre, which can send it to the next hospital	Must
CTD-09	See the hospital's answer and which ambulance it sent	Ambulance number and arrival time show on the case	Must
CTD-10	Move the case to another hospital, or ask for more than one ambulance	Every change saves who, when and why	Must
CTD-11	Show first-aid instructions in Arabic for the caller (CPR, bleeding, choking, birth)	The right instructions appear for the right problem	Should
CTD-12	Send the caller an SMS when the ambulance is on its way	SMS has the case number and arrival time	Could

4.2 Ambulance driver (AMD)

ID	What the system must do	How we check	Priority
AMD-01	Driver logs in to a shift on one ambulance	Jobs and tracking work only after login	Must
AMD-02	Driver gets a job with a sound alert and accepts it	Alert sounds even when the phone is locked	Must

ID	What the system must do	How we check	Priority
AMD-03	Map directions to the patient and to the hospital, also with no internet	Route shows with no network	Must
AMD-04	Send location every 10–15 seconds on a job and every 60 seconds when free; save it when there is no internet	After the network returns, the full route is there	Must
AMD-05	One button for each step: on the way, at the scene, leaving, at the hospital, free	Times in the case match the button presses (within 5 seconds)	Must
AMD-06	Mark the ambulance as not working, with a reason	The ambulance is no longer suggested for jobs	Should

4.3 Ambulance staff (AMS)

ID	What the system must do	How we check	Priority
AMS-01	See the job: address, how urgent, the problem, call notes	Works with no internet once received	Must
AMS-02	Find or add the patient, or add an "unknown patient"	Unknown patients get a temporary number to match later	Must
AMS-03	Enter vital signs many times; each entry saves the time and the paramedic's name	At least one entry before handover; strange values need a confirm	Must
AMS-04	Enter treatments and medicines given	Each one has time, dose and name	Must
AMS-05	Calculate a danger score from the vital signs	Patient cannot be marked "Stable" with a high score unless a reason is given	Should
AMS-06	Send a short patient summary to the hospital before arrival	Hospital gets it in 30 seconds with internet; it waits and sends later without	Must

ID	What the system must do	How we check	Priority
AMS-07	Record when a patient refuses help, with a signature or witness	Case cannot close as "Refused" without this form	Should
AMS-08	Close the case at the scene if the patient does not need the hospital	A reason is chosen from the closing list (section 5)	Must

4.4 Hospital emergency staff (HER)

ID	What the system must do	How we check	Priority
HER-01	Show patients coming in, sorted by priority then arrival time, with urgency and summary	List updates by itself; a sound plays for a new patient	Must
HER-02	Accept or refuse an incoming patient, with a reason	A refusal warns the dispatcher at once	Must
HER-03	Confirm the handover together with the ambulance crew	Handover time is saved; the ambulance becomes free	Must
HER-04	Update the emergency room status (open, limited, closed, cannot be reached) and free beds	If not updated for 4 hours, dispatchers see a warning	Must
HER-05	Start a hospital visit (tests and treatment) from the case	The visit is linked to the case and the patient	Must

New requirements because hospitals send their own ambulances:

ID	What the system must do	How we check	Priority
HER-06	Receive new requests from the Health Authority call centre with a sound alert; accept or refuse with a reason	A refusal or no answer in 5 minutes sends the case back to the call centre	Must

ID	What the system must do	How we check	Priority
HER-07	Choose one of the hospital's own free ambulances and send it the job	The job reaches the driver's app; if not confirmed in 60 seconds, ER staff phone the crew and record the call (time, who called, result) in the system	Must
HER-08	Add the hospital's own ambulances (one or more) and see them on a map with their status (free, busy, not working)	Positions update within 15 seconds	Should

4.5 Hospital staff (HST)

ID	What the system must do	How we check	Priority
HST-01	Search patients and open their record; allergies always shown at the top	Allergies show on every patient screen	Must
HST-02	Save the diagnosis with a standard disease code (ICD)	Cannot save without a code	Must
HST-03	Record when a patient goes home, moves to another hospital, or dies	This closes the emergency and frees the bed	Must
HST-04	Update free ward beds	Same 4-hour warning as HER-04	Must
HST-05	Tabs for medicines, lab tests, X-rays and scans, operations, vaccines, visits	Same as prototype screens 21–34, with errors fixed	Could
HST-06	Upload files, with limits on type and size and a virus check	Files that are too big or unsafe are refused with a message	Should
HST-07	Find and join double records of the same patient	Both histories are kept, the action is saved, and it can be undone	Must

New requirements for first-time and returning patients:

ID	What the system must do	How we check	Priority
HST-08	For a first-time patient, create a full new record: name, age, sex, phone, height, weight, blood type, allergies, long-term illnesses, current medicines, and a first health check (blood pressure, heart rate, temperature, oxygen, blood sugar, and the extra checks each hospital sets as recommended)	The record shows which required items are still missing; it cannot be marked complete until they are filled	Must
HST-09	For a returning patient, open the existing record and add only what is new for this visit	Search finds the patient before a new record can be created; no double records	Must

The paramedic can start a basic record at the scene (AMS-02); the hospital completes the full first-time record.

4.6 Admin (ADM)

ID	What the system must do	How we check	Priority
ADM-01	Add users, give them a role and limit them to a state, hospital or ambulance; login needs a second code sent by SMS	In tests, users cannot see data outside their limits	Must
ADM-02	Manage lists: areas and landmarks, hospitals, emergency types, urgency questions	Changes work without a new software release	Must
ADM-03	Manage ambulances (each linked to its hospital) and crew shifts	Each ambulance has a type, station and equipment	Must
ADM-04	Show performance numbers with a clear meaning for each, by state and date	The meaning shows when you point at a number	Must
ADM-05	See the record of who opened or changed what	Search by user, patient, case and date	Must

ID	What the system must do	How we check	Priority
ADM-06	Download reports to Excel or CSV	Downloads show only what the user is allowed to see	Should

5. Business rules

These rules apply everywhere. The system must enforce them; we do not rely on users to remember them.

ID	Rule
BR-01	Urgency levels: P1 = life in danger, P2 = serious, P3 = urgent but stable, P4 = not urgent. The Ministry of Health sets the meaning and the target arrival time for each level. [confirm]
BR-02	An ambulance can be sent only when the case has an urgency level, an area name, and a map point or landmark.
BR-03	A hospital can send only its own ambulances that are free and have a logged-in driver and crew. The call centre sees only hospitals that are working and have a free ambulance.
BR-04	Step times come from the button pressed in the app, not typed by hand. A manual change needs a reason and is saved.
BR-05	A closed hospital, or one that cannot be reached, cannot be chosen without a reason.
BR-06	Every case closes with one reason: taken to hospital, treated at the scene, refused, died at the scene, cancelled by caller, false alarm, patient not found, passed to another agency.
BR-07	A case cannot close as "taken to hospital" until the hospital confirms the handover.

ID	Rule
BR-08	Medical entries (vital signs, medicines, diagnosis) are never deleted. A correction makes a new version and keeps the old one.
BR-09	Every time someone opens a patient record, the system saves who and when. Emergency access to a record needs a reason.
BR-10	Times are saved in world time (UTC) and shown in Sudan time (UTC+2).
BR-11	Phone numbers are saved with the country code (+249).
BR-12	Each case number is unique across Sudan and starts with the state code, for example KH-2026-000123.

ID	Rule
BR-13	If a hospital refuses a case, or does not answer within 5 minutes, the call centre is warned and sends the case to the next nearest hospital.
BR-14	A patient seen for the first time gets one new record. The hospital must complete the full first-time record (height, weight, blood type and required health checks) before discharge.
BR-15	Before creating a new patient, staff must search first; the system warns if a similar patient already exists.

ID	Rule
BR-16	Priority is set by the system's built-in priority function from the triage answers, not by guesswork. Every list in every portal (call centre queue, hospital requests, incoming patients, ambulance jobs) is sorted by priority first, then by waiting time. If new information shows the patient is worse (for example a high danger score from the paramedic's vital signs), the system raises the priority and alerts the call centre and the hospital.

6. Data requirements

The system keeps twelve main types of information. "Required fields" must be filled in before saving.

Information	Required fields	Notes
Call	Caller number, start and end time, call-taker, result	Link to the call recording if calls are recorded
Case	Case number, urgency level, problem, area, map point or landmark, current step, closing reason	One case can have many patients and many ambulances
Patient	Name (Arabic and English), sex, age or date of birth, phone if known. First visit also: height, weight, blood type, allergies, long-term illnesses, current medicines	National number or driving licence if available; mother's name helps find the right person; temporary number for unknown patients
Vital signs	Time, entered by, heart rate, blood pressure, breathing rate, oxygen level, temperature, consciousness	Heart rhythm and blood sugar optional; many entries per patient
Treatment or medicine given	Time, entered by, name, dose, how given	Chosen from a Ministry-approved list

Information	Required fields	Notes
Ambulance	Plate number, type (basic or advanced), the hospital it belongs to, status	Equipment list
Location point	Ambulance, position, time, speed	A lot of data; kept for the agreed time
Hospital	Name, type, area, map position, emergency room status, total and free beds, last updated	One single list used on every screen
Hospital visit	Patient, hospital, start time, diagnosis code, how the visit ended	Linked to the case if it started as an emergency
Access record	User, time, action, record, reason	Can never be changed

New information for the hospital-based flow:

Information	Required fields	Notes
Hospital request	Case, hospital, time sent, answer (accepted or refused), answer time, reason if refused	One case can have several requests if hospitals refuse
First health check	Patient, date, height, weight, blood type, blood pressure, heart rate, temperature, oxygen, blood sugar, other tests ordered, entered by	Done once on the first visit; later visits add new checks

The prototype uses UAE addresses and +971 phone numbers. Replace them with Sudanese test data before testing.

7. Non-functional requirements

These describe how well the system must work: speed, safety, and staying on. The numbers are our proposal and become final when written in the contract.

ID	Area	What is needed	Target
NFR-01	Always on	Calls and dispatch work day and night	Working 99.9% of the month, plus a written paper-and-radio plan
NFR-02	Recovery	A second (backup) site	Back working within 1 hour; lose no more than 5 minutes of data
NFR-03	Speed	Case and dispatch screens	Open in under 2 seconds, even on slow internet (1 Mbps)
NFR-04	Load	Handle busy times	Tested at 3 times the busiest expected time [confirm call numbers]
NFR-05	No internet	Driver and paramedic apps keep working	Save at least 24 hours of data and send it when back online
NFR-06	Slow internet	Light screens	First screen under 500 KB; maps saved on the phone for Khartoum first, then other states
NFR-07	Security	Protect data and logins	Encrypted data, login with a second code by SMS, a security test every year
NFR-08	Access records	Save every view and change of patient data	Keep as long as the Ministry decides; we suggest 10 years [confirm]
NFR-09	Privacy	Collect only what is needed	A data rules document approved by the Ministry before launch
NFR-10	Hosting	Where the data is kept and who owns it	Amazon Web Services (AWS), nearest region; Health Authority approval for data outside Sudan [confirm]

ID	Area	What is needed	Target
NFR-11	Language	Arabic and English as equal languages; right-to-left for Arabic, left-to-right for English	All screens translated and checked by Sudanese users
NFR-12	Devices	Which devices are supported	Normal web browsers for computers; Android 9 or newer for the apps
NFR-13	Easy to use	Fast call handling	In tests, a P1 call is sent out in under 2 minutes (middle value)
NFR-14	Battery	Driver app with location on	Lasts a 12-hour shift with the phone charging in the vehicle
NFR-15	Backup	Copies of the data	Full copy every day; test bringing it back every 3 months
NFR-16	Watching the system	Warn the support team about problems	Alerts for system down, long call queues, and failed data sending

8. Integrations and standards

The first release needs four connections to other systems. The others come later. We use common international standards from the start, so adding them later is easy.

ID	Connects to	Why	When
INT-01	Phone system of the call centre	Receive emergency calls, show the caller's number, record calls, hold calls in a queue	First release
INT-02	SMS service (Zain)	Backup for sending jobs, messages to callers, links to share location	First release
INT-03	Maps (OpenStreetMap-based suggested)	Show maps, estimate travel time, save maps on phones	First release

ID	Connects to	Why	When
INT-04	Phone GPS	Live ambulance location from the driver app	First release
INT-05	DHIS2 (Ministry reporting system; no access yet)	Send monthly totals to the Ministry	Later
INT-06	National ID system	Check who the patient is	Later
INT-07	Hospital lab and X-ray systems	Bring test results and images into the record	Later
INT-08	Police and civil defence	Share cases that need them	Later

Standards: HL7 FHIR R4 to share health data; ICD-10 or ICD-11 codes for diseases [confirm]; LOINC codes for lab tests; the Ministry's essential medicines list; international time format (UTC); international phone format (+249).

9. Assumptions, open items and sign-off

Everything marked [confirm] in this document must be answered before the build starts. These are the main questions:

- [ ] Which emergency number will the public call? (Today 333 is the known ambulance number and 999 is police. The telecom regulator, TPRA, must confirm it. The system will let admins set the number, so it can change later.)
- [ ] Confirmed: the Sudan Health Authority call centre answers calls and sends each request to a nearby hospital; the hospital sends its own ambulance.
- [x] ~~Decided: no pilot. How many calls, ambulances and hospitals does Khartoum have, so we can size the system?~~
- [ ] What do P1 to P4 mean, and what is the target arrival time for each? (Ministry of Health to decide.)
- [ ] Who writes the urgency questions? (Ministry doctors.)
- [ ] Decided: hosted on AWS. Still open: who owns the data, and is storing it outside Sudan approved?
- [ ] How long do we keep access records, and is the data rules document approved?
- [ ] ICD-10 or ICD-11 for disease codes?

- [ ] Which ambulance services are included: Ministry only, or also Red Crescent, NGOs and private?
- [x] ~~Decided: each hospital has one or more ambulances and adds them in the system.~~
- [ ] Which health checks are required for every first-time patient, and which only when the doctor orders them?

Sign-off: